

1. AI Surveillance Drone





Autonomous Surveillance Drones

Ground Vehicles for Threat Neutralization



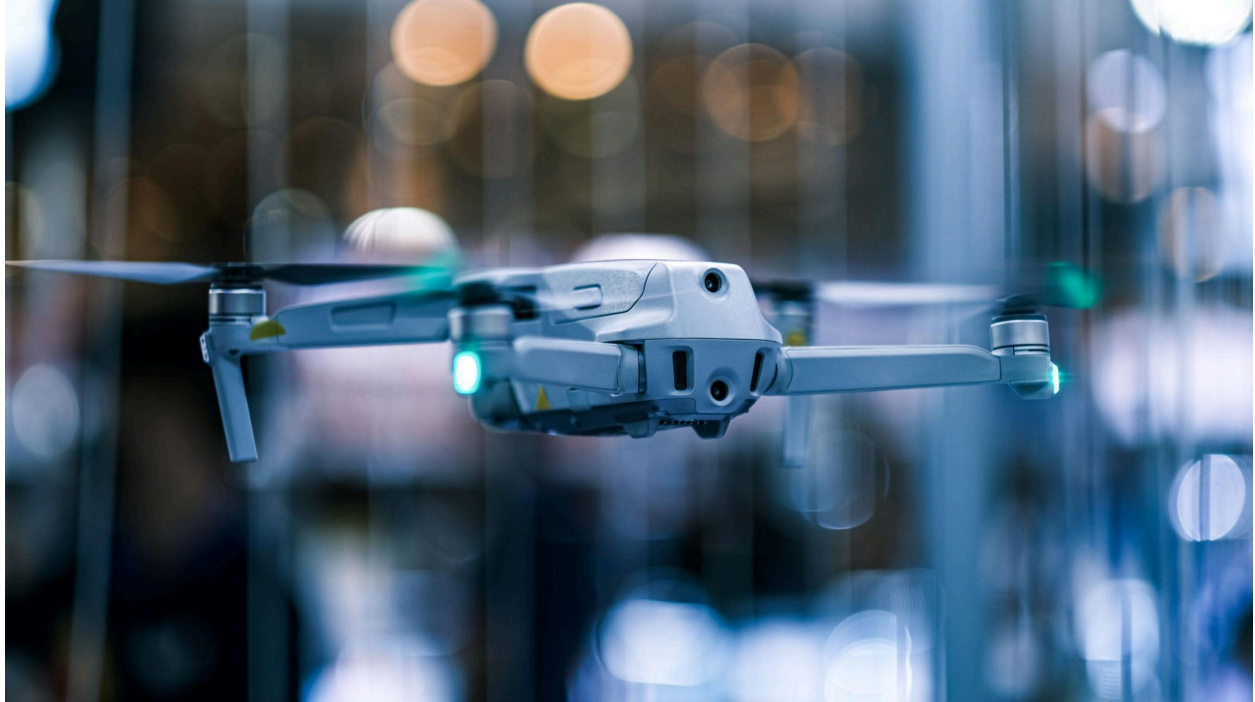
 Indeema

Source: indeema.com









Hardware Stack

- Jetson Nano
- Thermal Camera
- GPS Module
- IMU Sensor
- LoRa Module
- Flight Controller
- Brushless Motors
- LiPo Battery

What It Can Do

- Human detection
- Night surveillance
- Autonomous patrol
- Live video transmission
- GPS waypoint navigation

AI Features

- YOLO object detection
- Thermal target detection

- Intruder tracking
- AI-based alerts

Software Stack

- Ubuntu Linux
- Python
- OpenCV
- TensorFlow/PyTorch
- MAVLink
- ROS

Career Domains

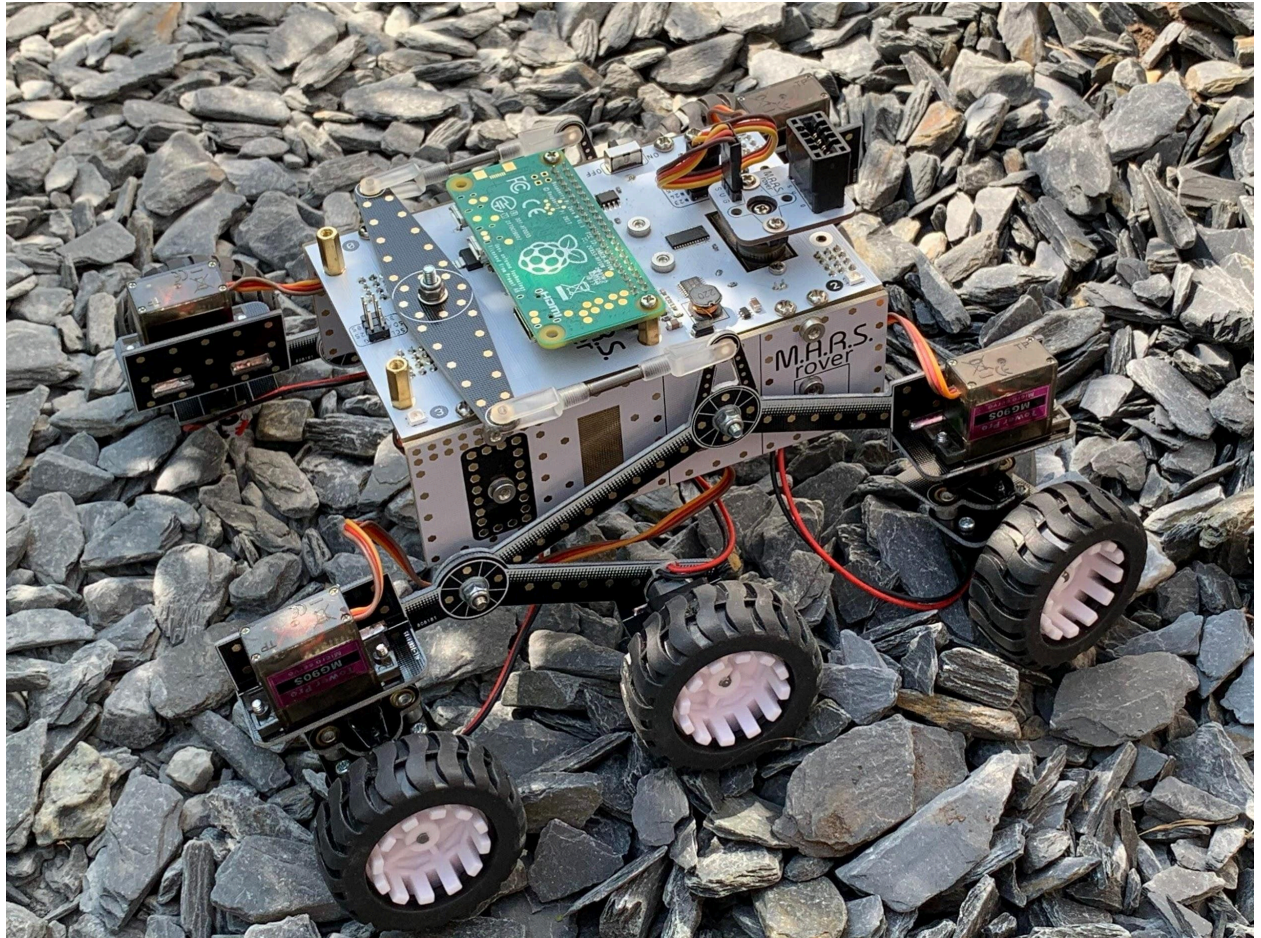
- Defense
 - Aerospace
 - Surveillance
 - Autonomous UAVs
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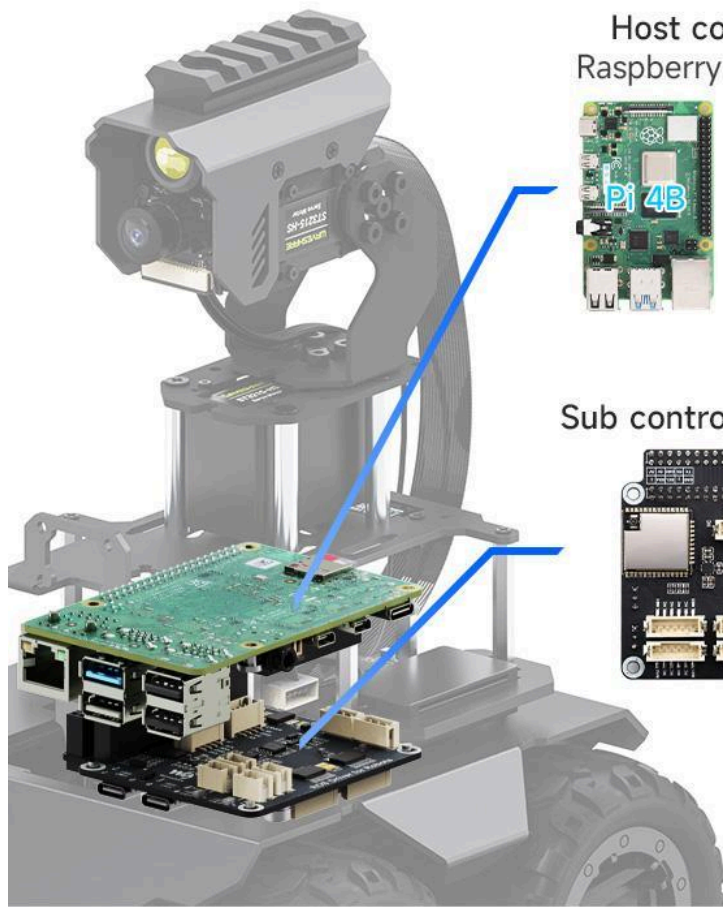
2. Autonomous Rover











Host controller:
Raspberry Pi 4B/Pi 5



Advanced AI
Computer Vision
Flask Web Application
Real-time video
Audio Interaction
Jupyter Lab Server
...

Sub controller: ESP32



High-Frequency PID
Controller
Wheel Linear Speed
Closed-Loop Control
Pan-tilt Servo Control
And Feedback
LED Light ON/OFF
OLED Screen
IMU Data Reading
Battery Voltage Monitoring
...



STM32 Vision Smart Car

Visual Recognition | Ultrasonic Obstacle Avoidance | Infrared Line Tracking



Six-Line Following



Five-Line Following



ROS

Hardware Stack

- Raspberry Pi
- STM32
- GPS Module
- Ultrasonic Sensors
- Motor Driver
- Camera Module
- IMU Sensor

Features

- Obstacle avoidance

- Autonomous navigation
- GPS tracking
- Path planning
- Remote control mode

AI Functions

- Lane detection
- Object recognition
- Voice commands
- Mapping

Technologies

- ROS
- OpenCV
- Python
- Embedded C

Real Applications

- Delivery robots
 - Military rovers
 - Warehouse robots
 - Agricultural robots
-

3. Smart Defense System

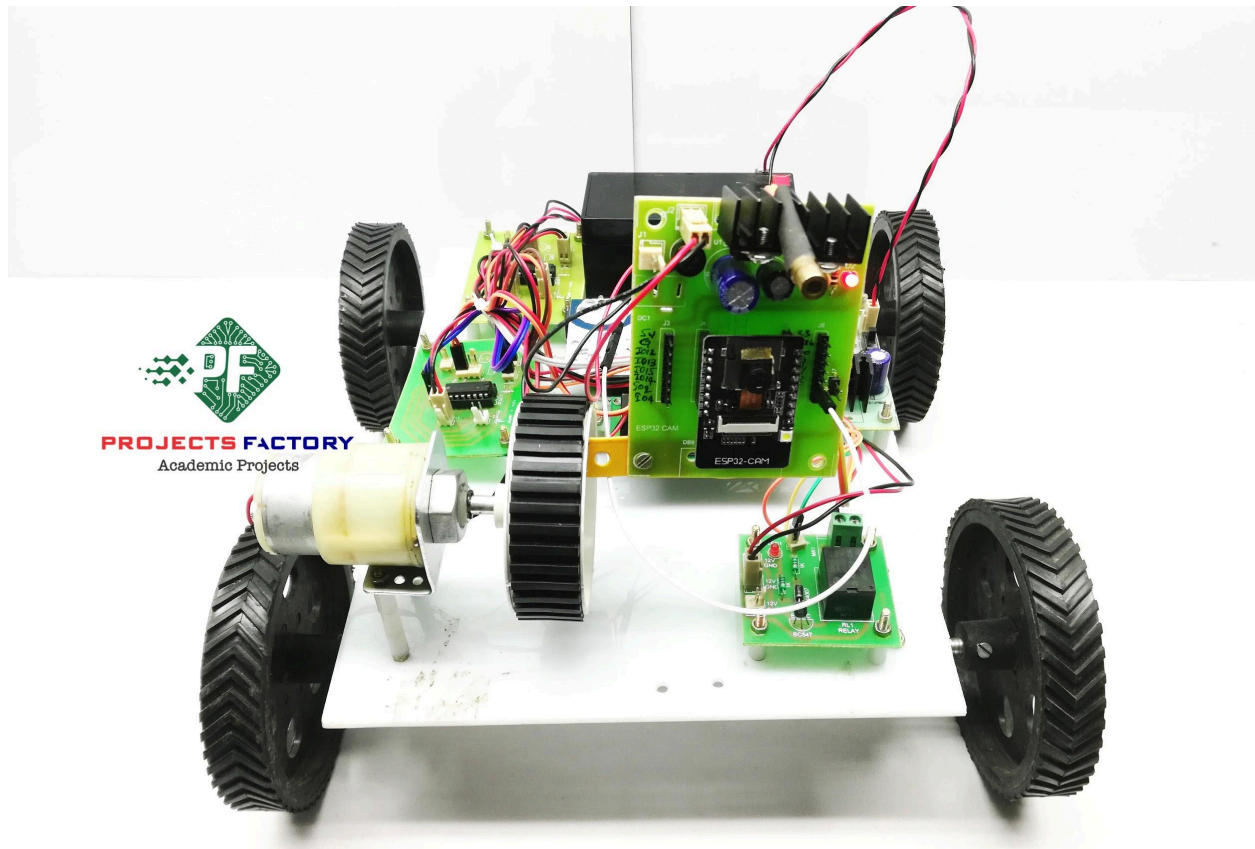












Hardware Stack

- ESP32
- Thermal Camera
- LoRa Module
- PIR Sensors
- Camera Module
- Alarm System

Main Features

- Intruder detection
- Long-range alerts
- Thermal monitoring
- Wireless surveillance
- AI target classification

AI Features

- Human detection
- Heat signature analysis
- Smart threat alerts

Communication

- LoRa mesh network
- Wi-Fi
- MQTT

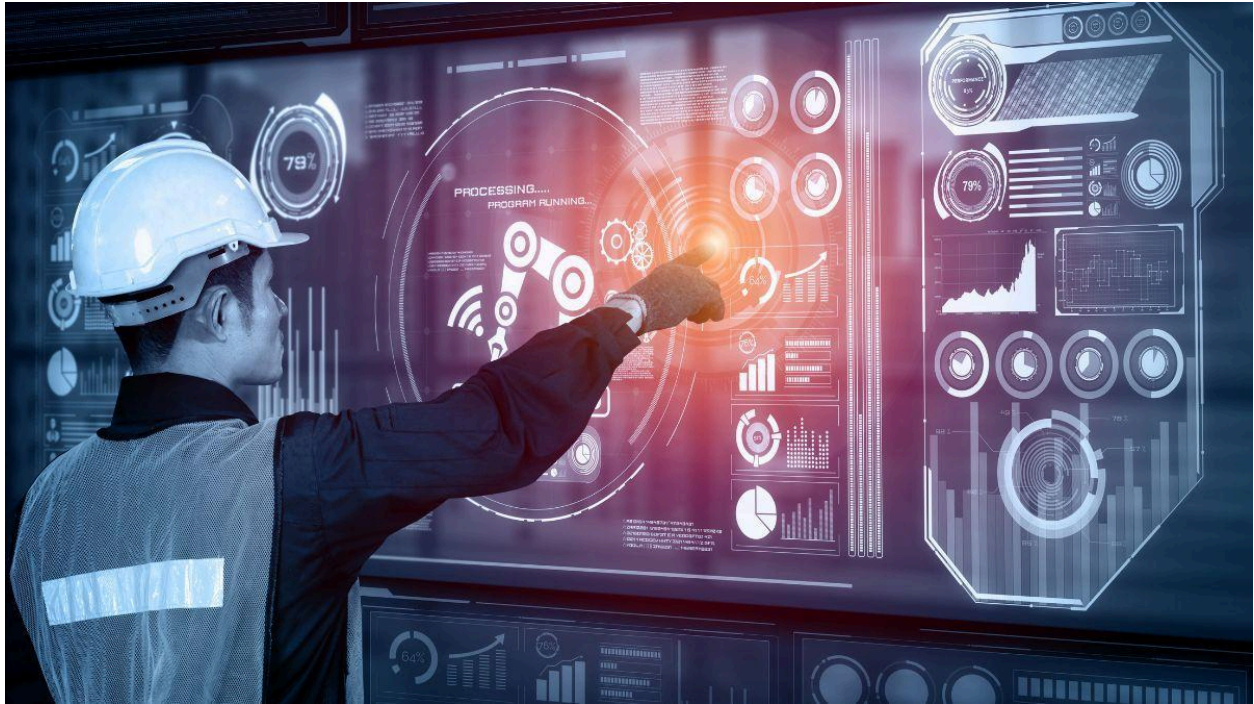
Use Cases

- Border security
 - Forest surveillance
 - Restricted area monitoring
 - Smart military systems
-

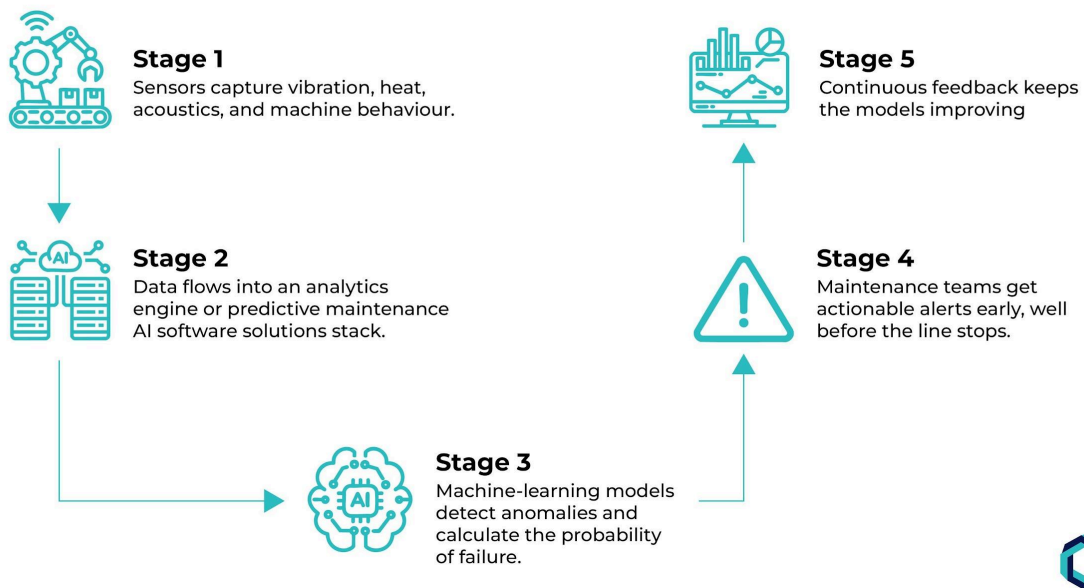
4. Industrial AI Monitoring







How Do AI-Predictive Maintenance Services Work?



Hardware Stack

- STM32

- Raspberry Pi
- Temperature Sensors
- Vibration Sensors
- Gas Sensors
- Cloud Dashboard

Features

- Machine health monitoring
- Predictive maintenance
- Fault detection
- Real-time analytics
- Remote monitoring

AI Features

- Failure prediction
- Anomaly detection
- Smart alerts
- Data analysis

Cloud Technologies

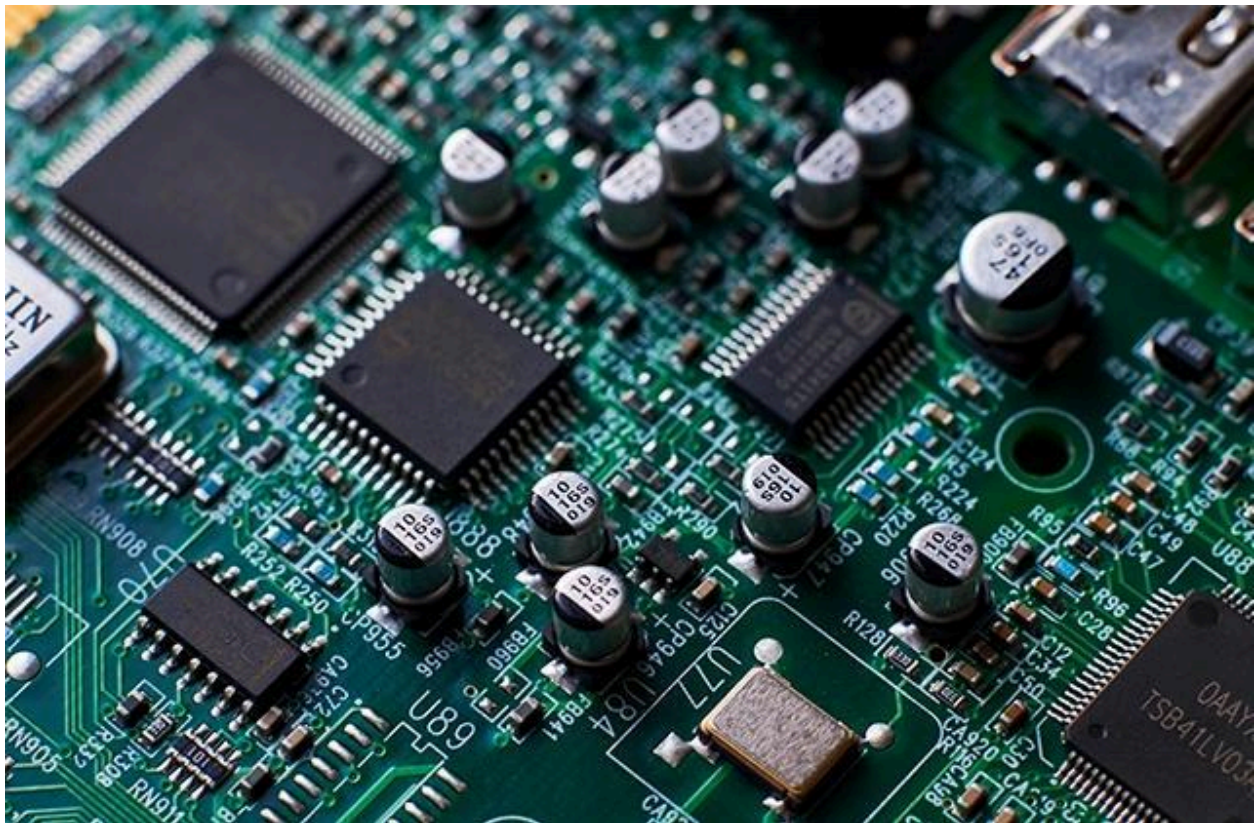
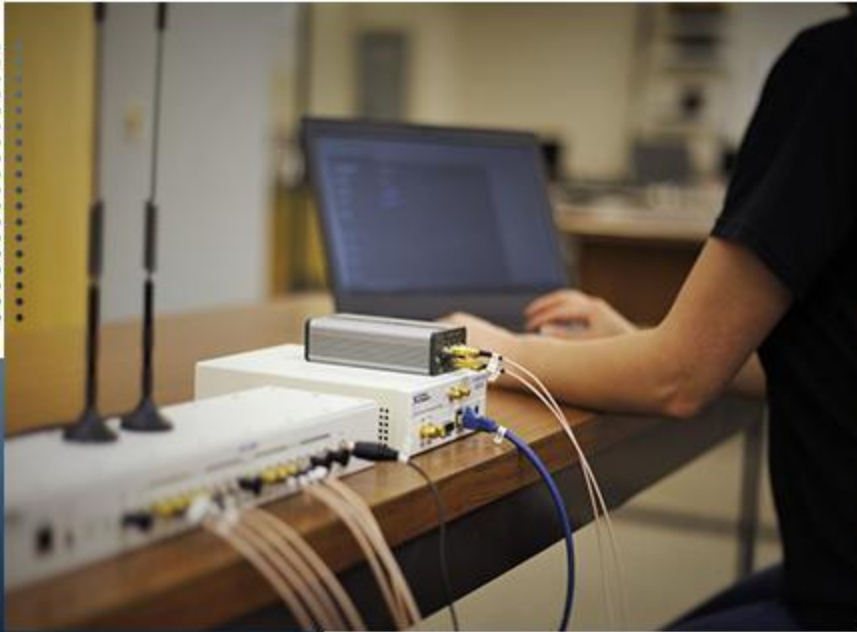
- MQTT
- Node-RED
- Grafana
- Firebase

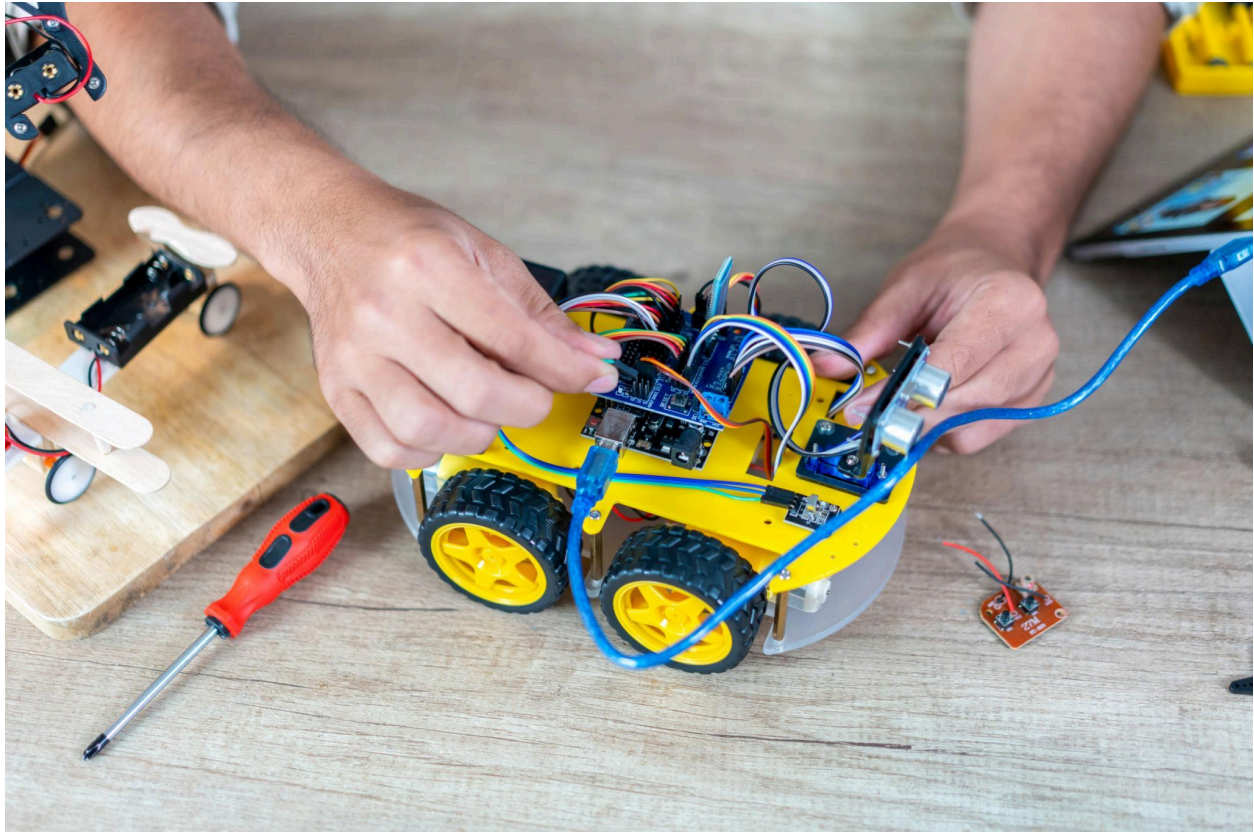
Industries

- Manufacturing
- Power plants
- Automation
- Smart factories

Skill Matrix Needed







Domain	Skills
Embedded	ESP32, STM32, Raspberry Pi
AI	OpenCV, YOLO, TensorFlow
Robotics	ROS, Motor Control
Networking	LoRa, MQTT, Wi-Fi
Hardware	Sensors, PCB Design
Linux	Ubuntu, Shell Commands

Recommended Learning Order

Step 1

- C Programming
- Electronics basics
- Arduino/ESP32

Step 2

- Raspberry Pi
- Linux
- Python

Step 3

- Sensors + Communication
- LoRa
- MQTT

Step 4

- OpenCV
- AI object detection
- YOLO

Step 5

- Robotics
- ROS
- Autonomous navigation

Ultimate Goal Projects

Project	Difficulty
Smart Helmet	Beginner
LoRa Communication	Beginner
Autonomous Rover	Intermediate
AI CCTV	Intermediate

Industrial AI Monitoring Advanced

Smart Defense System Advanced

AI Surveillance Drone Expert

Best Future Career Path

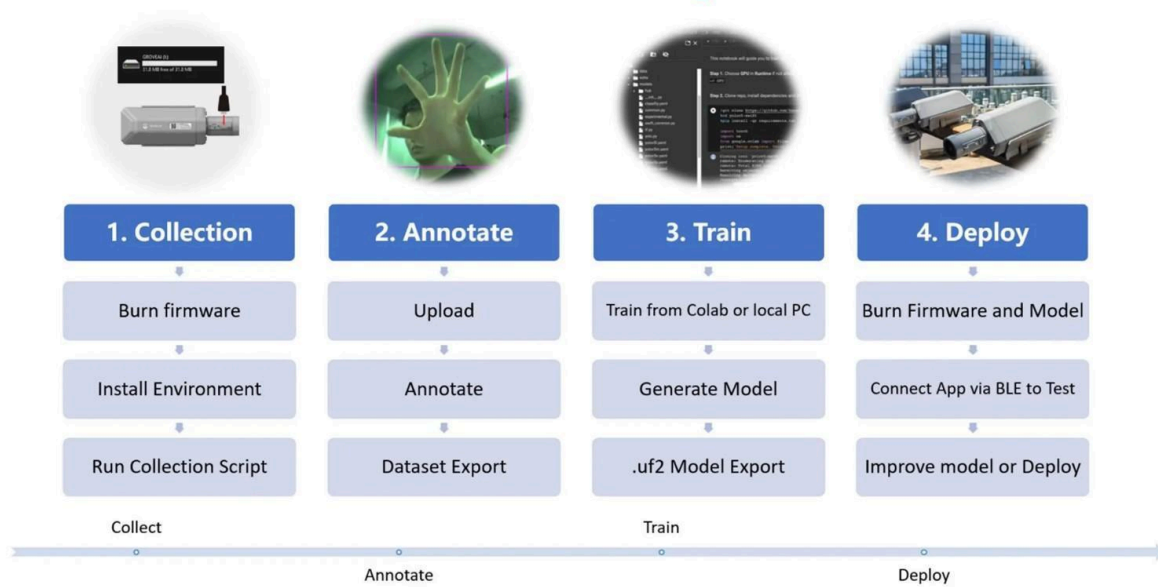






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AI Model Training







- Embedded AI Engineer
- Robotics Engineer
- Autonomous Systems Engineer
- Defense Technology Engineer
- Aerospace Systems Engineer
- Industrial Automation Engineer